



NC STATE
UNIVERSITY

NRP 
Nuclear Reactor Program

NE235 Nuclear Reactor Operations Module #8

Reactor Period, Inhour Equation, and Exponential Power Equations

Dept. of Nuclear Engineering
North Carolina State University

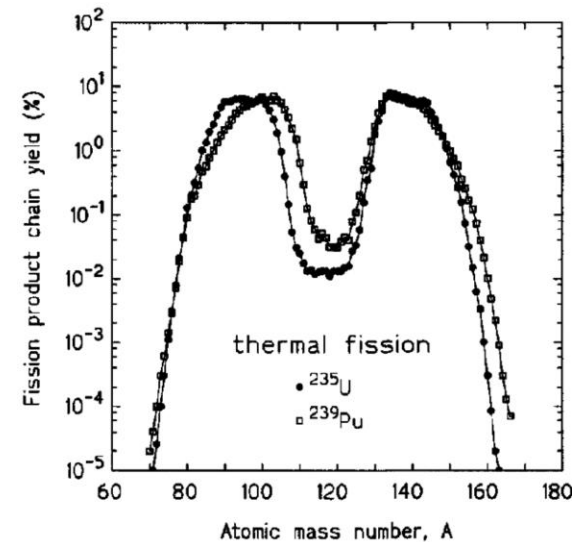
NE235 Nuclear Reactor Operations Training

Module #8

Module #8 Contents:

1. Inhour Equation
2. Inhour Plots: Reactivity ρ –vs- Reactor Period τ
3. One Group Approximation for Small Reactivity Insertions
4. Exponential Power Equations

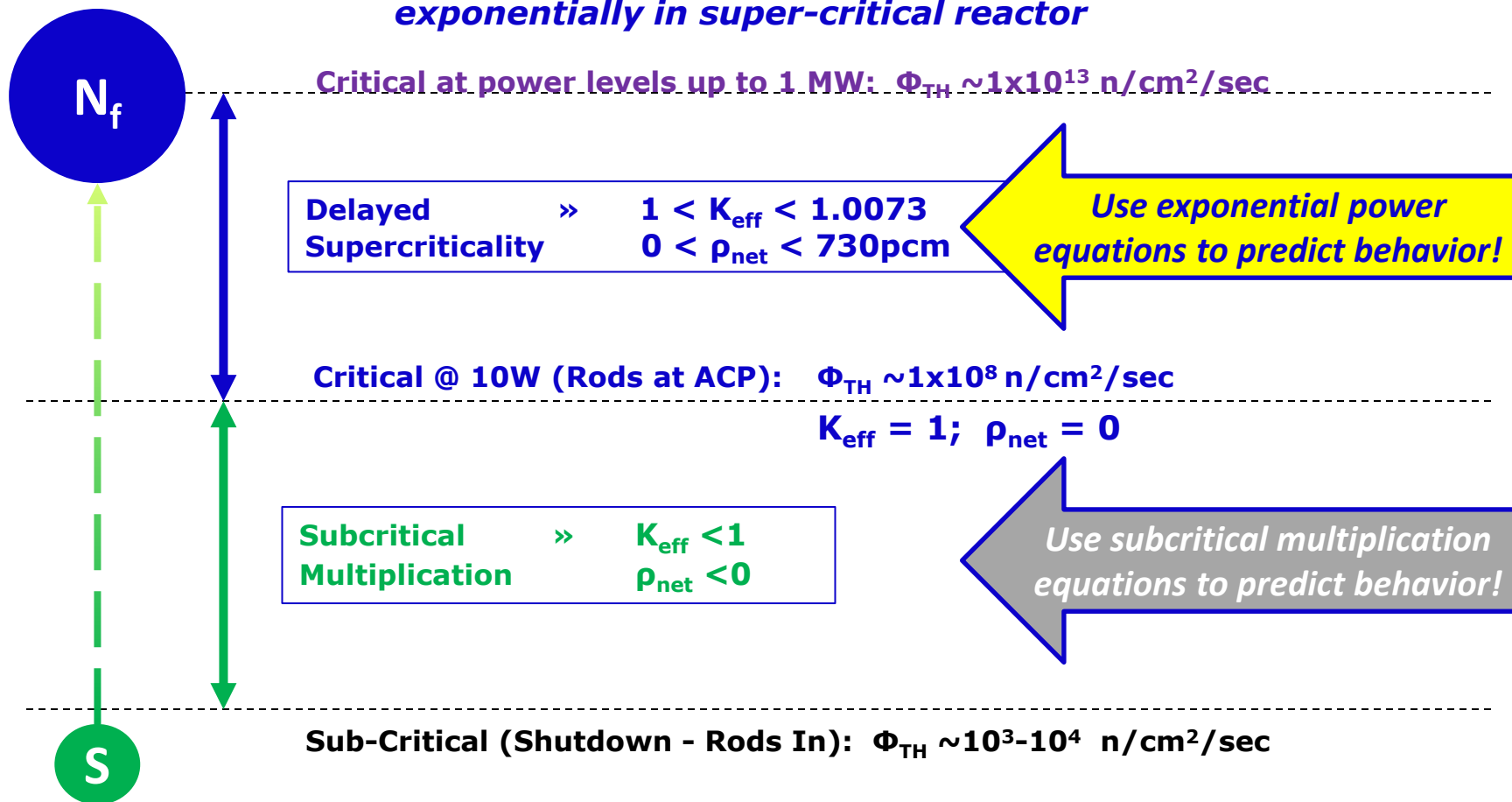
Homework #8 – Due 11/1/2022



Shultis, J.K and Faw, R. E, *Fundamentals of Nuclear Science and Engineering*, Pg. 145

RECAP: Subcriticality –vs- Supercriticality

Fission Neutrons; N_f - Increase through subcritical multiplication, then exponentially in super-critical reactor



Source Neutrons; $S \sim$ Constant for each Start-Up

Definitions

Reactor Period (τ):

≡ Time, in seconds, to change power by a factor of $e = 2.718$.

“SUR” Start-Up Rate in Decades per Minute (DPM):

A 1 DPM SUR => Reactor power is increasing by 1 decade (x10) per minute

A 2 DPM SUR => Reactor power is increasing by 2 decades (x100) per minute

“Inhour” Unit:

The amount of reactivity (pcm or $\frac{\Delta k}{k}$) yielding a +1 hour (3600 sec) period

Inhour Equation:

Reactivity ρ –vs- Reactor Period τ

The rate of power change (expressed as period or startup rate) is related to reactivity by the “Inhour” equation:

$$\rho = \frac{\ell^*}{k\tau} + \sum_{i=1}^6 \left(\frac{\beta_i}{1 + \lambda_i \tau} \right) \quad \text{Eq\# 2.9}$$

Where:

ℓ^* = prompt neutron lifetime (sec)

τ_{sec} = Period (sec)

β_i = Fractional yields for 6 groups of delayed neutron precursors.

λ_i = Decay constants for 6 groups of delayed neutron precursors.

Therefore, the “Inhour” equation relates Period τ_{sec} to $\rho \left(\frac{\Delta k}{k} \right)$

Delayed Neutron Pre-Cursors – 6 Groups

TABLE 3.5*

 Delayed neutron data for thermal fission in ^{235}U

Group	Half-life sec	Decay constant $\lambda_i, \text{sec}^{-1}$	Energy keV	Yield, neutrons per fission	Fraction β_i
1	55.72 (80.4)	0.0124	250	0.00052	0.000215
2	22.72 (32.8)	0.0305	560	0.00346	0.001424
3	6.22 (8.97)	0.111	405	0.00310	0.001274
4	2.30 (3.32)	0.301	450	0.00624	0.002568
5	0.610 (0.380)	1.14	—	0.00182	0.000748
6	0.230 (0.332)	3.01	—	0.00066	0.000273

WEIGHTED AVERAGE = 9.03 (13.0) - SEC
 ↑ MEAN LIFE
 ↘ MEAN HALF LIFE

Total yield: 0.0158
 Total delayed fraction (β): 0.0065

*Based in part on G. R. Keepin, *Physics of Nuclear Kinetics*, Reading, Mass.: Addison-Wesley, 1965.

 Decay Constants λ_i

 Yield Fractions β_i

Inhour Equation:

Reactivity ρ/β –vs- Reactor Period τ

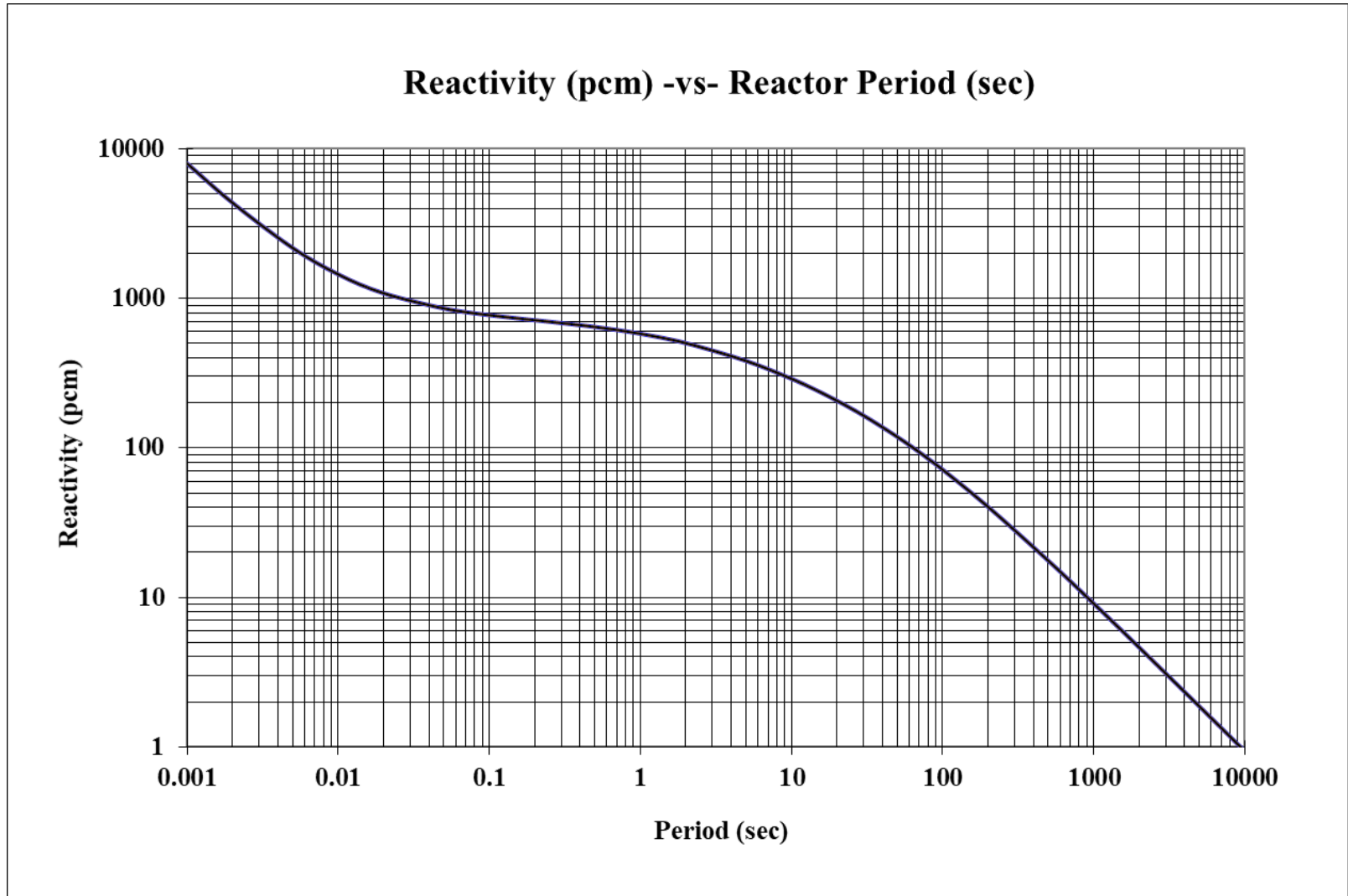
Since β_i and β_{eff} vary through core life we divide Inhour by β to yield reactivity in dollars $\$(\rho/\beta)$:

$$\frac{\rho}{\beta} = \frac{\ell^*}{k\tau\beta} + \sum_{i=1}^6 \left(\frac{\beta_i}{\beta(1 + \lambda_i\tau)} \right)$$

The advantage of using dollars is it always correctly defines the boundary for prompt criticality. If $\rho/\beta = \$1$, $\rho = 730\text{pcm}$, which is the boundary for prompt criticality, or the reactor state of being supercritical without having to wait for delayed neutrons.

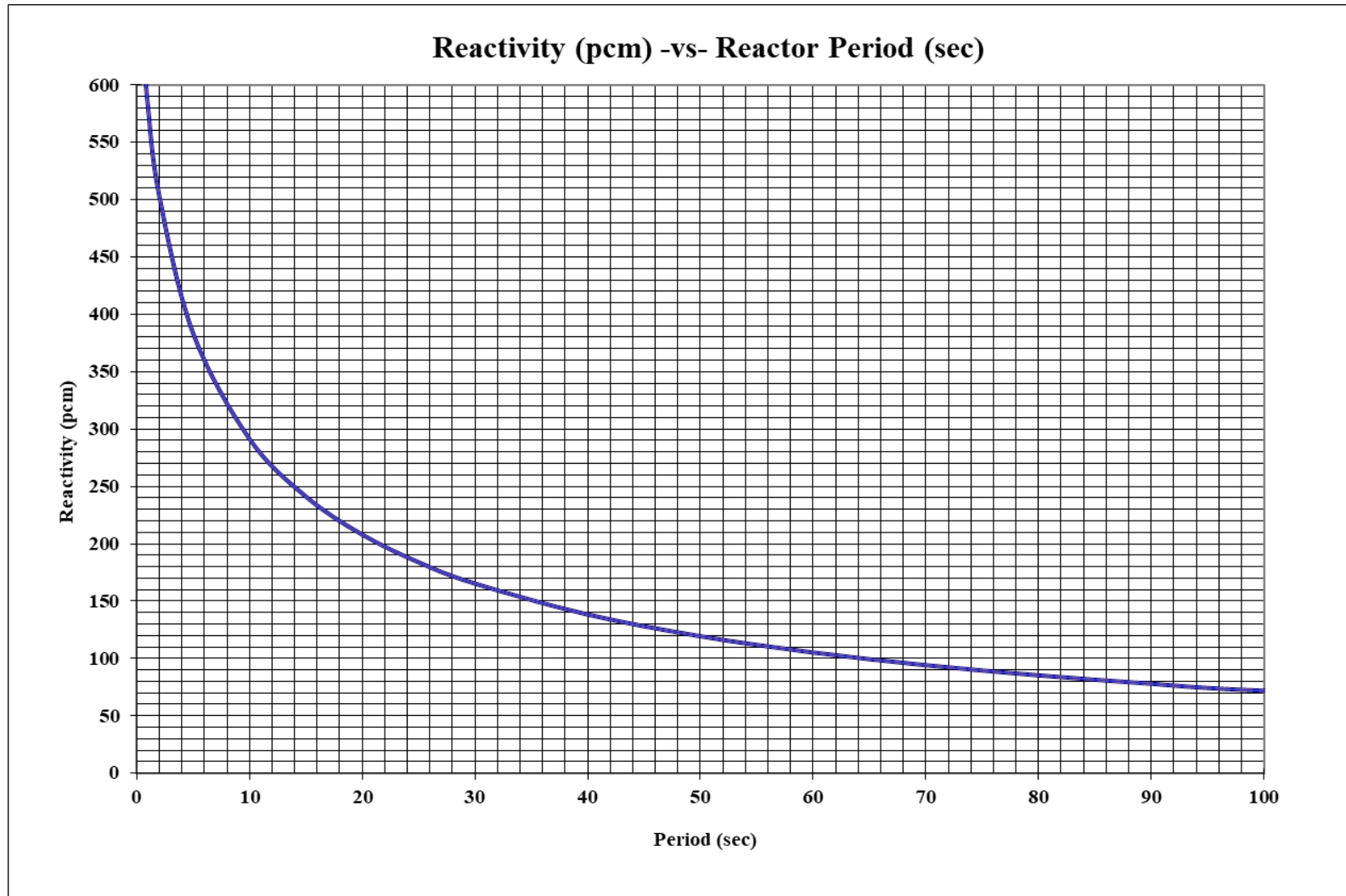
Inhour Plot*:

Reactivity (pcm) – vs- Period (sec) (log/log)



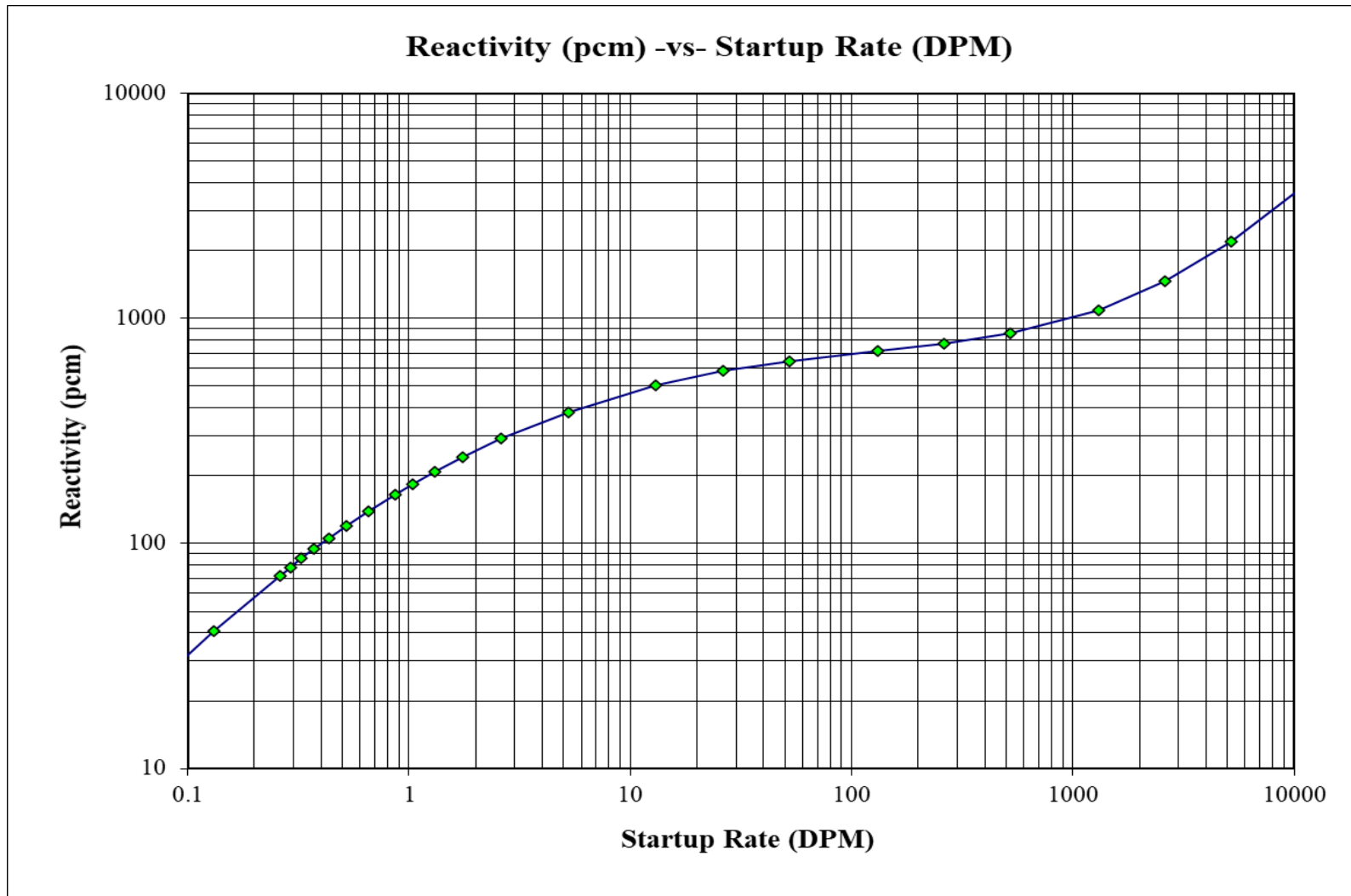
Inhour Plot*:

Reactivity (pcm) – vs- Period (sec) (lin/lin)



Inhour Plot*:

Reactivity (pcm) –vs- Startup Rate (DPM)



One-Group Approximations to InHour Eq.

Approximations:

1. For short periods (**where $\rho \gtrsim \beta$**) the delayed neutron term drops out and:

$$\rho = \underbrace{\frac{\ell^*}{K\tau}}_{\text{Prompt Neutron Term}} + 1$$

2. For long periods (**where $\rho \ll \beta$**) the prompt neutron term drops out and we have the “One Group Approximation for Small Reactivity Insertion”:

$$\rho = \frac{\beta_{eff}}{1 + \lambda_{eff}\tau} \quad \text{Eq\# 2.10}$$

Where $\lambda_{eff} \sim 0.1$ (1/sec) for slightly supercritical

Note: **ρ** and **β** must be in the same units (**pcm vs $\Delta K/K$**)!!

Approximations -vs- Graph Results

Example 1:

If $\rho_{\text{net}} = +100 \text{ pcm}$, what is the reactor period τ ?

- One Group Approximation for Small Reactivity Insertion:

$$\rho = \frac{\beta}{1+\lambda\tau} \therefore \tau = \frac{\left(\frac{\beta}{\rho}-1\right)}{\lambda} = \frac{\left(\frac{730}{100}-1\right)}{0.1\left(\frac{1}{\text{sec}}\right)} = 63 \text{ sec}$$

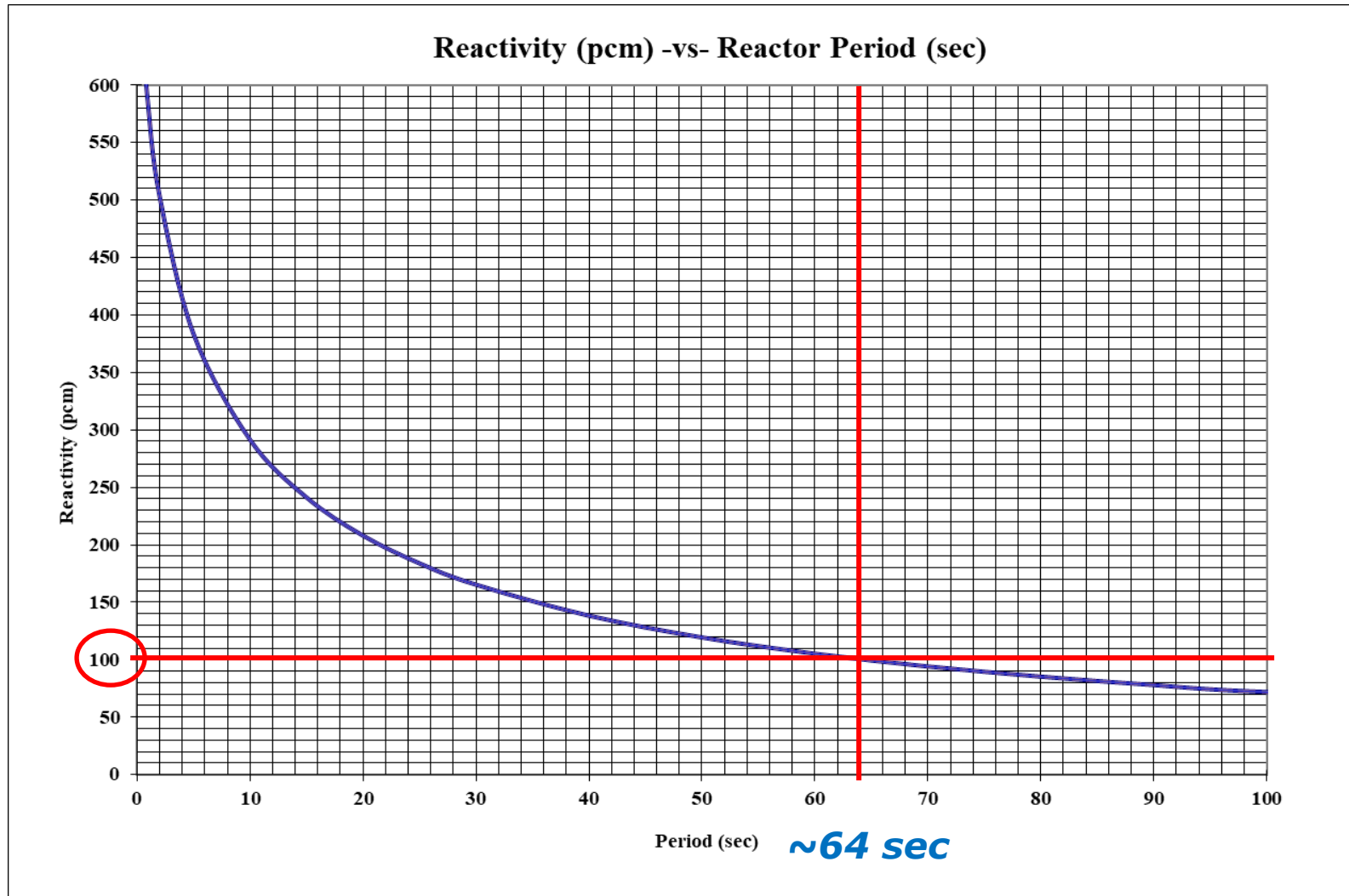
- From Graph (linear/Linear) – *see next page*:

$$\tau = 64 \text{ sec}$$

So, good agreement between graph and approximation for small reactivity.

Inhour Plot*:

Reactivity (pcm) – vs- Period (sec) (lin/lin)



Approximations vs Graph Results

Example 2:

If $\rho_{\text{net}} = +500 \text{ pcm}$, what is the period τ ?

- One Group Approximation:

$$\rho = \frac{\beta}{1+\lambda\tau} \therefore \tau = \frac{\left(\frac{\beta}{\rho}-1\right)}{\lambda} = \frac{\left(\frac{730}{500}-1\right)}{0.1\left(\frac{1}{\text{sec}}\right)} = 4.6 \text{ sec}$$

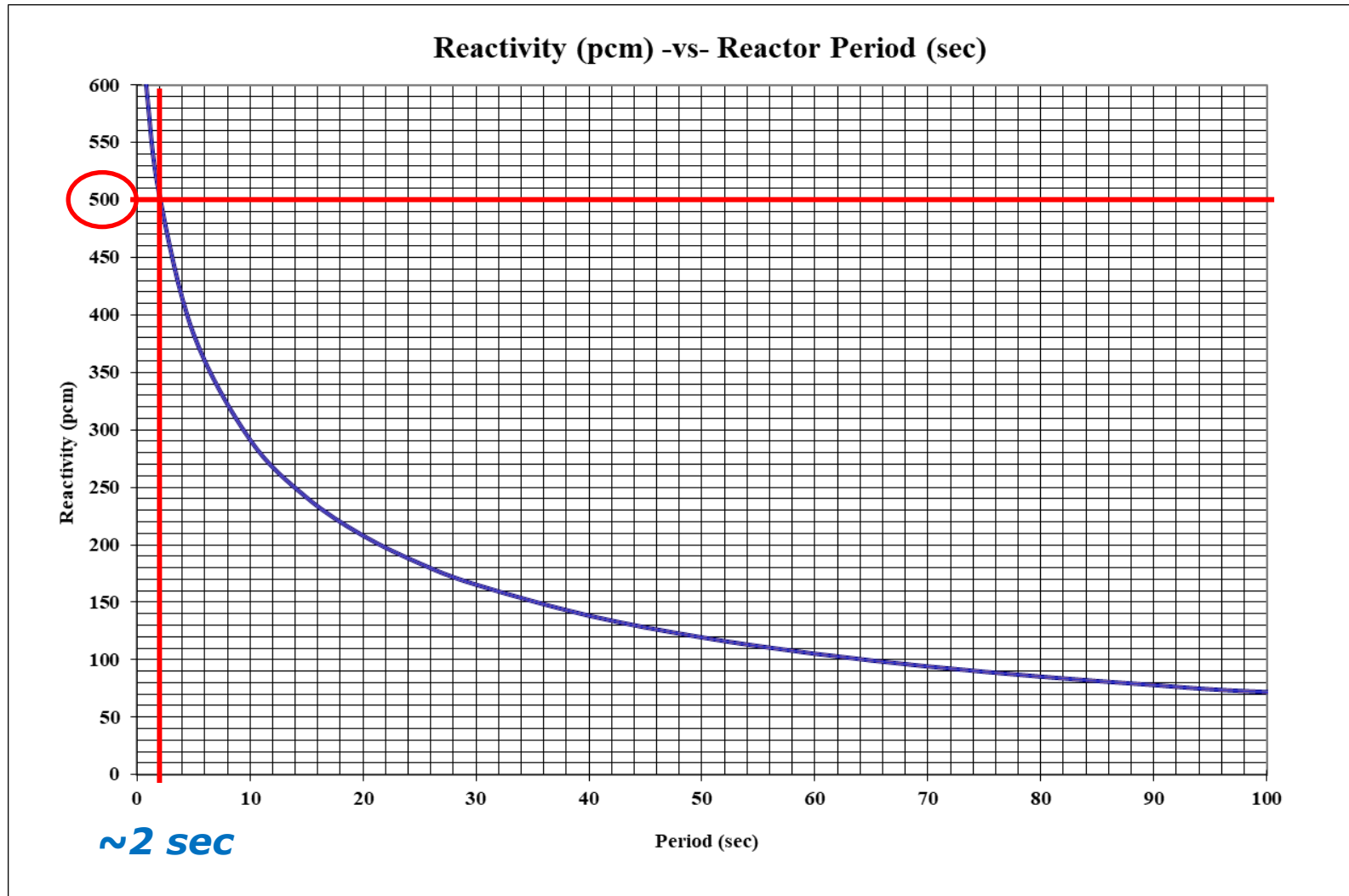
- From Graph – *see next page*:

$$\tau = 2 \text{ sec}$$

So, poor agreement between graph and approximation for larger ρ_{net} .

Inhour Plot*:

Reactivity (pcm) – vs- Period (sec) (lin/lin)



Exponential Power Equations

$$\left. \begin{aligned} P(t) &= P_0 e^{t/\tau} \\ P(t) &= P_0 2^{t/dt} \\ P(t) &= P_0 10^{t/DT} \\ P(t) &= P_0 10^{SUR \cdot t} \end{aligned} \right\} \text{Eq. 2.5 \& 2.6}$$

Where:

$P(t)$ = power as a function of time.

P_0 = initial power

t = Time over which power is changing (sec)

τ = period (sec)

DT = Decade time (minutes) $\equiv$ Time to increase 1 decade (x10) where
($DT = 1/SUR$)

dt = doubling time (sec) $\equiv$ Time to double reactor power

SUR = Startup Rate (# decades per minute power is increasing) (DPM)

Rate Conversions

$$\tau \text{ (sec)} = 1.44 \times dt \text{ (sec)}$$

$$\text{SUR}(DPM) = \frac{26.1}{\tau \text{ (sec)}}$$

$$\text{SUR}(DPM) = \frac{18.1}{dt \text{ (sec)}}$$

$$DT \text{ (min)} = 1 / \text{SUR} \text{ (DPM)}$$

Eq. 2.7

Exponential Power Equations

Example:

How long will it take to go from **10 Watts** to **10 Kilowatts** if

$\rho_{\text{net}} = +100 \text{ pcm}$?

From previous example, **$+100 \text{ pcm} \Rightarrow \tau = 64 \text{ sec}$** from Inhour curve

So, if:

$$P(t) = P_0 e^{t/\tau}$$

$$10,000 = 10 e^{t/64}$$

$$\ln\left(\frac{10,000}{10}\right) = \frac{t}{64}$$

$$\therefore t = 442 \text{ sec}$$

Exponential Power Equations

Example – *continued*:

And if $\rho_{\text{net}} = +500$, $\tau = 2$ sec from Inhour curve

$$P(t) = P_0 e^{t/\tau}$$

$$10,000 = 10 e^{t/2}$$

$$\ln\left(\frac{10,000}{10}\right) = \frac{t}{2}$$

$\therefore t = 14$ sec to go from 10 W to 10 kW

Applications - Relating time to reactivity via Inhour Eq. allows us to:

- a) Perform control rod calibrations.**
- b) Measure reactivity worth of insertions.**
- c) Predict the rate of power increase.**

Lab #5 – Control Rod Calibration

Objective: Measure the reactivity –vs- length for a reactor control rod.

Laboratory Procedure Snapshot:

Withdraw rod, measure power doubling time (dt), calculate reactivity worth of pull via Inhour, and plot differential & integral rod worth –vs- control rod height.

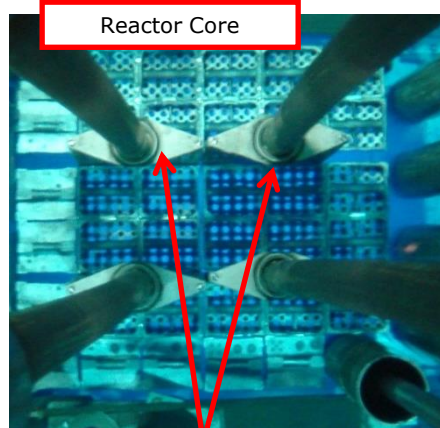
Reactor Theory concepts:

- K_{eff} –vs- Reactivity ρ
- InHour Equation: Reactivity ρ –vs- Period (τ)
- Core Neutron Flux Distribution

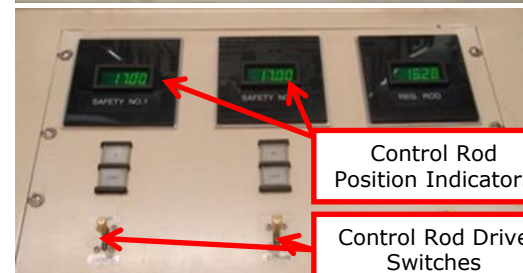
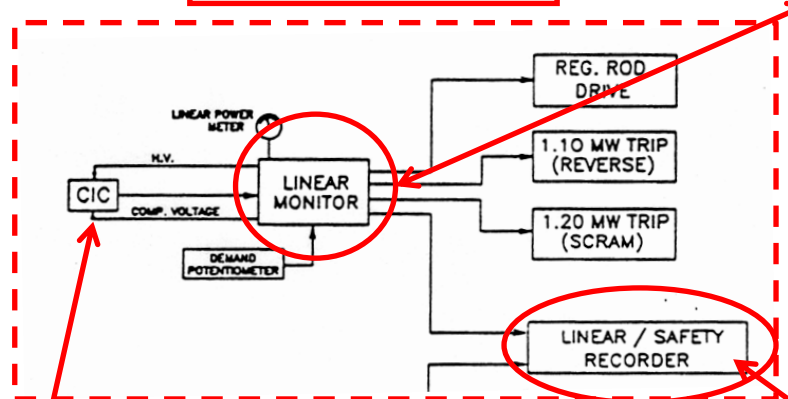
Physical Plant Elements:

- Control Rods
- Linear Channel and chart recorder

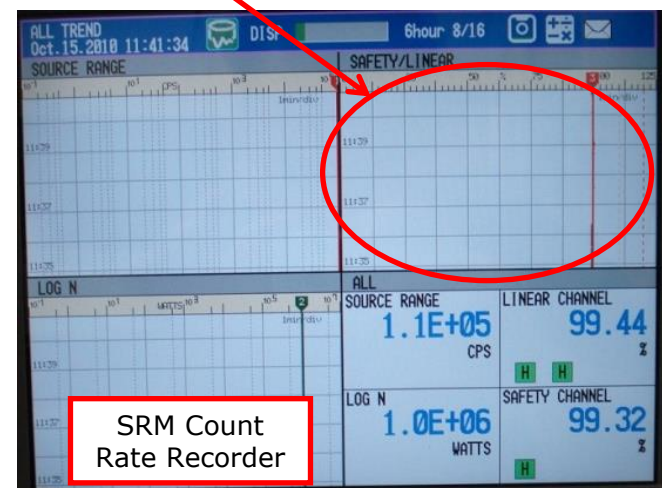
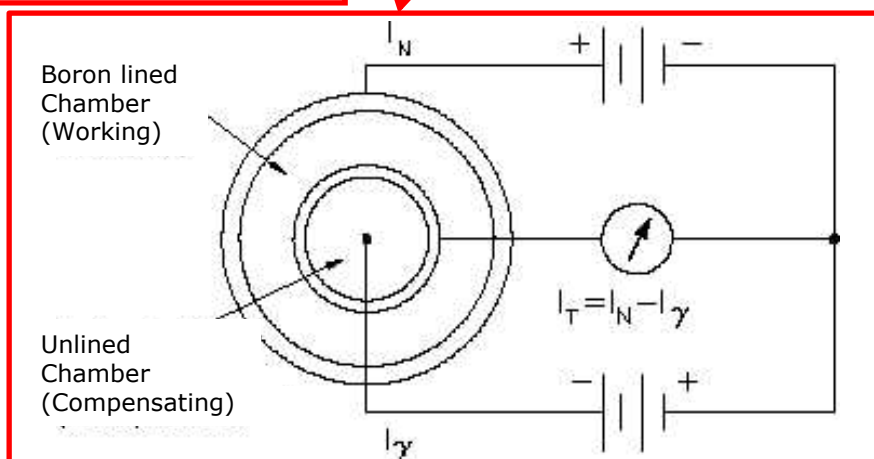
PULSTAR Reactor: Instrumentation for Control Rod Calibration



Linear Power Channel Schematic



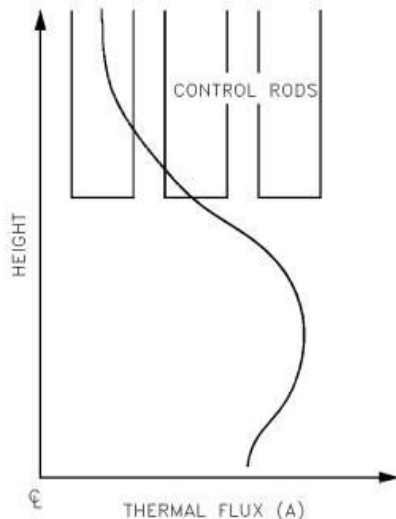
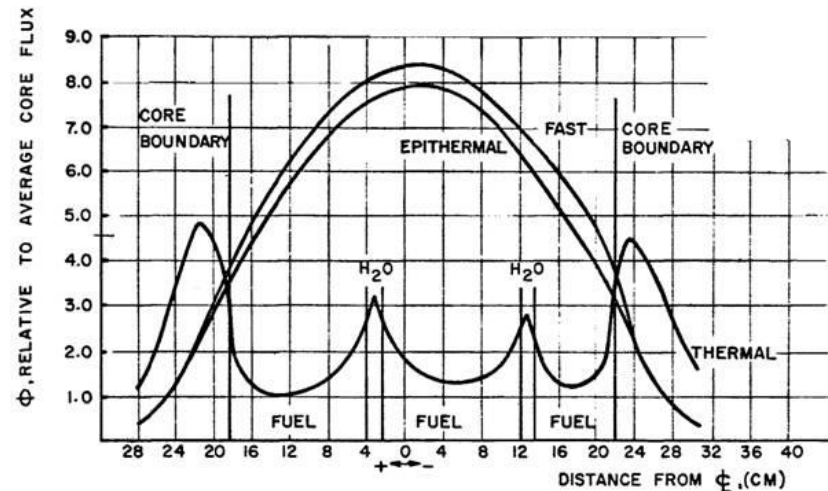
Compensated Neutron Ion Chamber Schematic



Reactor Theory Concepts: Core Neutron Flux Distribution

Neutron Flux Distribution:

- Control rod 'worth' per unit length is a function of thermal neutron flux distribution in the core.
- Vertical and horizontal flux gradients in core due to leakage at core edges
- Local flux gradients in proximity to control rods, water gaps, instrumentation penetrations, etc...

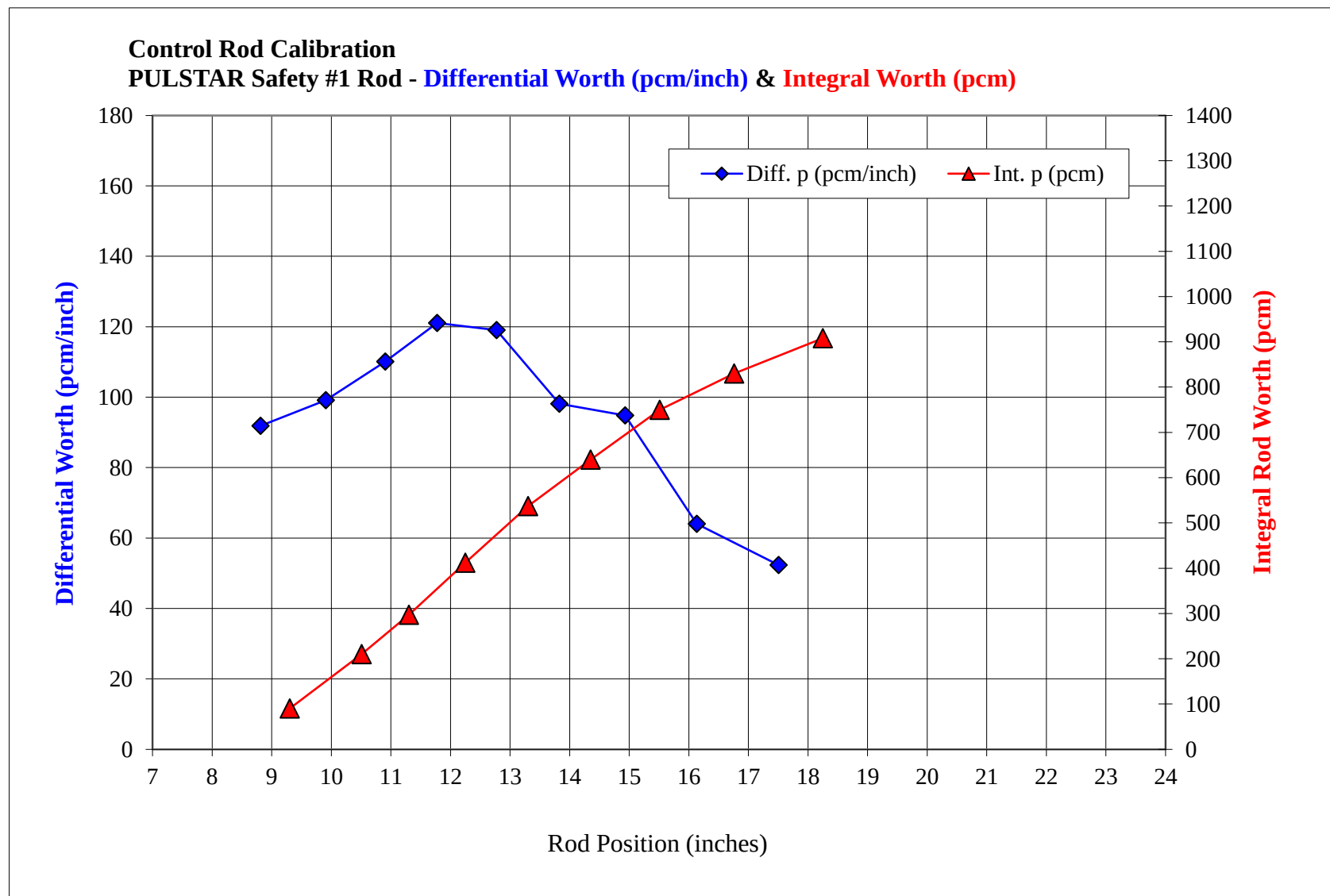


Control Rod Calibration:

Measure Differential Worth (pcm/inch), and Integral Worth (pcm):

- Bank rods so that the rod to be calibrated is fully inserted.
- Withdraw rod in steps, measuring time to double reactor power (doubling time dt (sec)).
- Convert dt to period ($\tau = 1.44 \times dt$), and period to reactivity ρ using InHour curves.
- Plot Differential (pcm/inch) and Integral Rod (pcm) Worths -vs- Rod Height.

Safety #1 Rod Calibration Data



Summary

- ❑ **Inhour Equation:** *Relates reactivity to rate of change of reactor power, commonly expressed as reactor period.*
- ❑ **Inhour Plots:** *Expresses Reactivity ρ –vs- Reactor Period t , or –vs- reactor Startup Rate (DPM)*
- ❑ **One Group Approximation for Small Reactivity Insertions:** *Convenient approximation for calculating relationship between reactivity and period for small reactivity values.*
- ❑ **Exponential Power Equations:** *Determine power level as a function of time given the rate of change of reactor power expressed as τ , dt , DT , or SUR .*